Object-Conditioned Strategy-Level Joint Parameter-Bundle Configuration for Haptic Teleoperation

Fengjie Ma[a], Chuang Cao[b], Hua Zhang[a,*], Qixin Cao[b], Yilin Zhou[a]

[a] *School of Mechanical and Automotive Engineering, Shanghai University of Engineering Science, 333 Longteng Road, Shanghai, 201620, China*
[b] *School of Mechanical Engineering, Shanghai Jiao Tong University, 800 Dongchuan Road, Shanghai, 200240, China*

Abstract

Fixed teleoperation settings are difficult to tune across objects with different fragility, geometry, and grasping requirements. Most related object-conditioned approaches focus on slave-side impedance, whereas coordinated assignment of the slave response, operator-facing haptic interface, and gripper execution has received less attention. This paper presents an object-conditioned, strategy-layer framework that maps each detected object class to one of three operational strategies and atomically assigns a joint parameter bundle to these three subsystems. The contribution is a constrained systems architecture: low-rate visual perception is asynchronously integrated with a nominal 200-Hz supervisory loop, the first threshold-qualified mapped detection initiates a one-shot configuration event, and strategy locking prevents repeated intra-trial switching. Three operators completed 27 matched task blocks comprising 135 grasp-and-transfer trials with six objects and five control modes. Relative to impedance-only reconfiguration, joint parameter-bundle configuration reduced mean completion time by 1.79 s (8.5%; matched-block bootstrap 95% CI, 1.10–2.51 s). The mean difference favored the joint parameter bundle for each operator and object; descriptively, this mode also achieved the highest observed success rate and the lowest Raw NASA-TLX score. These findings describe a within-sample system-level time advantage under the tested conditions; they do not identify the separate effects of the haptic-interface and gripper settings. Because visual-event histories and synchronized trigger–contact timestamps were not retained during formal trials, the study does not verify trial-level trigger correctness or pre-contact completion of configuration.

Keywords: haptic teleoperation, impedance control, telemanipulation, object-conditioned parameter configuration, strategy-level joint parameter-bundle configuration, human-in-the-loop evaluation

1. Introduction

Teleoperation leverages human judgment to perform remote manipulation tasks that remain difficult for autonomous robots, particularly in hazardous environments, flexible manufacturing, and unstructured settings. Haptic feedback conveys remote contact information to the operator and supports assessment of contact state, grasp stability, and slip. Previous studies have examined the stability and transparency of bilateral teleoperation and the adaptation of controllers to the environment, operator, and task [1, 2]. In practical mechatronic systems, however, bilateral control must operate together with perception, mechanical execution, and the human–machine interface.

For heterogeneous objects, a fixed parameter set must trade off gentle contact against positioning and transfer stability. Lower stiffness and grasp force can reduce contact severity but may impair positioning and holding stability, whereas higher settings can improve positional robustness at the cost of greater impact or deformation risk. These competing requirements motivate object-conditioned configuration across multiple subsystems rather than a single fixed setting.

Impedance control regulates the relationship among displacement, velocity, and interaction force, allowing a robot to remain compliant during contact [3]. Fixed impedance is straightforward to implement but inevitably represents a compromise across objects. Variable-impedance methods instead adapt stiffness and damping according to task state, contact information, or learned policies, while accounting for the stability implications of time-varying parameters [4, 5]. Passivity-preserving variable-stiffness control has also been addressed with energy-tank formulations that explicitly manage energy exchanged during stiffness variation [6]. Previous teleoperation and tele-impedance studies have extensively regulated remote-robot stiffness and impedance according to operator motion, estimated human impedance, contact state, or task phase [7–13]. Bilateral force feedback and operator-facing haptic feedback have also been integrated with impedance-regulated teleoperation [9, 14–16]. These studies show that remote mechanical compliance and operator-facing feedback can be integrated within teleoperation systems, while their primary emphasis is generally placed on impedance transmission, stiffness regulation, transparency, or feedback presentation.

Visual and multimodal information has subsequently been used to support semi-autonomous impedance adaptation. A vision-and-voice tele-impedance method selected slave-side impedance from object properties while retaining an operator

*Corresponding author
Email address: huazhang@yeah.net (Hua Zhang)

confirmation or correction step [17], and visual-impedance control has combined visual and force information in a visual-feature space [18]. At the workflow level, intention inference and task-oriented shared control have reduced manual decision-making [19]. More recent approaches have derived stiffness from object geometry, material, and environmental relationships [20] or generated task-relevant stiffness matrices from visual, gaze, and language inputs [21]. Thus, visual object attributes, geometry and material, task intent, and related multimodal inputs have primarily been translated into remote stiffness, impedance matrices, or control-authority decisions.

Related studies on automatic switching and passive arbitration have addressed how control authority is allocated and when it changes [22, 23], and haptic interfaces have been evaluated for telemanipulation performance [14]. Teleoperated grasping work has also used tactile deformation and grip-safety information to regulate gripper impedance for maintaining grip force [24]. These lines of work demonstrate the value of object- and task-dependent configuration, but they typically address the remote arm, the operator interface, or the gripper as separate design elements. Among the closely related studies reviewed here, we did not identify a framework in which one detected object-semantic condition directly assigns a common parameter bundle spanning slave-arm impedance, master-side haptic rendering, and gripper execution. The unresolved system-level question is therefore not whether vision can regulate impedance or whether haptic and gripper parameters can be adapted individually, but whether a single interpretable operational strategy can configure these three subsystems jointly and yield task-level benefits beyond visual-semantic display, manual strategy selection, or impedance-only reconfiguration.

To address this question, we propose an object-conditioned strategy-layer configuration architecture designed for contact-preparatory parameter setting in haptic teleoperation of heterogeneous objects. Each detected object class is mapped to an operational strategy and then to a joint parameter vector. After task initiation, the first threshold-qualified mapped detection initiates a single configuration transaction, and the selected strategy is locked for the remainder of the trial. The asynchronous implementation keeps low-rate perception outside the nominal supervisory call path and avoids continuous online optimization. Because operator motion was not gated by perception and synchronized trigger–contact timestamps were not recorded, trial-level contact-preparatory timing could not be verified.

The central positioning of this work is an interpretable, strategy-layer configuration architecture rather than post-contact continuous optimization or persistent visual servoing. Object class serves as a task prior for a coordinated, atomic assignment of the slave-side mechanical response, the operator-facing haptic interface, and gripper execution. This assignment creates a consistent operating state across three otherwise separate mechatronic subsystems; it is not a dynamically derived coupling law, impedance-matching relationship, or statistically demonstrated synergy among channels. The architectural contribution lies in combining semantic strategy selection, bounded asynchronous queues, non-blocking supervisory polling, one-shot event triggering, and intra-trial strategy locking into a constrained configuration workflow. It is therefore a system-level configuration architecture and experimental evaluation, rather than a new impedance controller or visual detector. The formal study evaluates this architecture in a closed set of six known object instances; it does not establish generalization to unseen objects or verify trial-level trigger correctness.

The system-level evaluation uses five modes: fixed-parameter baseline (A), manual-selection workflow baseline (B), joint parameter-bundle configuration (C), visual-semantic display with fixed parameters (D), and impedance-only reconfiguration (E). The evaluation focuses primarily on whether the complete parameter bundle improves task completion time relative to impedance-only reconfiguration under matched visual-semantic and impedance conditions. The C–D comparison assesses joint parameter-bundle configuration beyond the visual-semantic display, while trajectory length, pause count, success rate, and workload are interpreted as secondary or exploratory outcomes. Operator- and object-stratified results describe whether this difference has a consistent direction; they do not identify the independent causal contributions of the haptic and gripper parameter groups.

The main contributions are as follows:

1. **Object-conditioned strategy-layer parameter-bundle architecture.** The framework elevates object class from a display cue to a strategy-layer event that atomically assigns a seven-dimensional parameter bundle across slave mechanical response, operator-facing haptic rendering, and gripper execution. Object classes are grouped into fragility-priority, balanced, and stability-priority operational strategies and mapped to operating points constrained by hardware capabilities, task risks, and preliminary screening. The contribution is consistent configuration across subsystems, not a dynamic coupling model between channels.
2. **Asynchronous, event-triggered one-shot reconfiguration.** Image acquisition at 15 fps and independent YOLO11n inference communicate with a nominal 200-Hz supervisory update through bounded queues. Non-blocking polling isolates perception from the supervisory call path; one-shot triggering converts semantic output into a single configuration transaction; strategy locking avoids repeated intra-trial switching caused by detection fluctuations; and an approximately 300-ms transition smooths changes in stiffness and the controller damping-scaling parameter.
3. **Human-in-the-loop system-level evaluation across five modes.** Five modes were implemented on an Omega.7–Panda platform to distinguish visual-semantic display with fixed parameters, a manual-selection workflow baseline, impedance-only reconfiguration, and joint parameter-bundle configuration. The primary C–E comparison evaluates the complete parameter bundle at the system level and does not separate the individual effects of the added haptic-interface and gripper channels. The evaluation involved three operators, six objects, and 135 trials; its inferential scope is limited to the repeated measurements

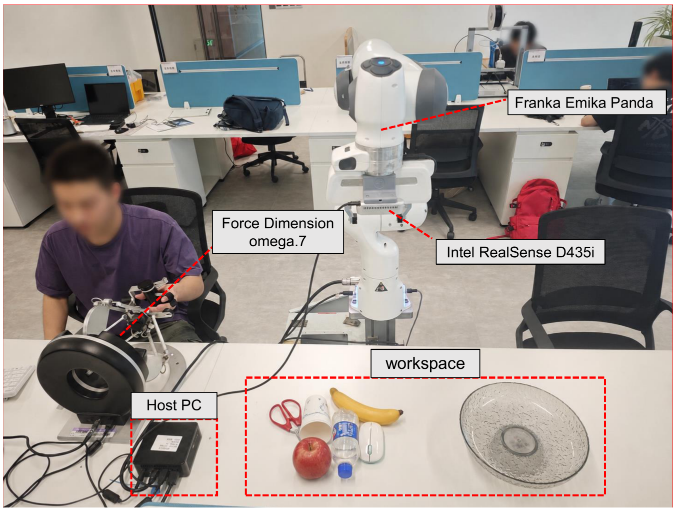

Figure 1: Experimental teleoperation platform showing the Omega.7 master device, Franka Panda robot, Intel RealSense D435i camera, host computer, and object workspace. The Franka Hand gripper is mounted at the Panda end effector.

in this sample.

2. Methods and System Implementation

2.1. Mechatronic System Architecture and Asynchronous Execution

The experimental platform comprised an Omega.7 force-feedback master device, a seven-degree-of-freedom Franka Panda robot, a Franka Hand gripper [25], an Intel RealSense D435i camera, and a control computer (Fig. 1). The D435i provided 424 × 240-pixel color images at 15 fps; the depth stream was not used. The control computer ran visual detection, the nominal 200-Hz supervisory teleoperation update, parameter configuration, haptic rendering, gripper control, and data logging.

Figure 2 illustrates the timing relationships within this multi-rate asynchronous software architecture.

RGB frames were acquired every 66.7 ms and written to a single-slot queue, preventing stale images from accumulating. YOLO11n ran in an independent process and returned class labels and confidence scores through a two-slot result queue. The nominal 200-Hz supervisory loop polled this queue every 5 ms without waiting for inference, allowing camera acquisition, perception, and operator motion to proceed in parallel. Per-image processing time was evaluated separately from the 15-fps camera sampling period, as reported in Section 4.1. Table 1 summarizes the update timing, triggers, and data flow of the system modules.

2.2. Master–Slave Mapping and Implementation of the Mechatronic Channels

Only translational motion of the Omega.7 was mapped to the Panda. The master-device position increment and desired slave position were updated as

$$\Delta\mathbf{p}_m(k) = \mathbf{p}_m(k) - \mathbf{p}_m(k-1), \tag{1}$$

$$\mathbf{p}_d(k) = \mathbf{p}_d(k-1) + S\,\mathbf{C}\Delta\mathbf{p}_m(k), \tag{2}$$

where the position-scaling factor was $S = 3.0$ and $\mathbf{C} = \mathrm{diag}(-1, -1, 1)$. The desired end-effector orientation $\mathbf{R}_d$ was fixed at its prescribed value throughout each trial; active rotational input from the operator was not mapped. At a conceptual level, Cartesian impedance control acted on the six-dimensional pose error,

$$\mathbf{w}_c = \mathbf{K}(c)\mathbf{e} + \mathbf{D}(c)\dot{\mathbf{e}}, \tag{3}$$

$$\mathbf{K}(c) = \mathrm{diag}(K_t, K_t, K_t, K_r, K_r, K_r), \tag{4}$$

where $\mathbf{e}$ contains translational error relative to $\mathbf{p}_d$ and rotational error relative to the fixed $\mathbf{R}_d$. Thus, K_r changes resistance to rotational disturbances and the ability to hold the prescribed orientation; it does not scale operator-commanded rotation. Here $\mathbf{D}$ denotes the controller's effective damping action rather than an independently identified physical Cartesian damping matrix.

The slave-controller API internally generated its normalized damping action from the active diagonal stiffness matrix and a scalar API parameter denoted by ζ. In the implementation convention,

$$\mathbf{D}_{\mathrm{API}}(c) = 2\zeta(c)\sqrt{\mathbf{K}(c)}, \tag{5}$$

where the square root is applied elementwise to the diagonal entries. $\mathbf{D}_{\mathrm{API}}$ denotes the API's normalized internal construction and is not reported as a physical damping matrix in SI units. Accordingly, ζ is treated only as the dimensionless damping-scaling parameter exposed by the controller API and is not interpreted as a Cartesian critical-damping ratio, because no effective Cartesian mass or inertia matrix was identified.

For haptic rendering, the Franka internal state estimator provided the slave-side external-force estimate $\mathbf{F}_{\mathrm{ext}}$ in newtons. The baseline master-side force command along direction i was

$$u_{h,i}^{\mathrm{base}} = \mathrm{sgn}\big(K_f F_{\mathrm{ext},i}\big)\max\big(\big|K_f F_{\mathrm{ext},i}\big| - d, 0\big), \quad i \in \{x, y, z\}, \tag{6}$$

where K_f is a dimensionless force-scaling factor and d is a per-axis dead-zone threshold in newtons.

A separate positive-z cue was generated solely from the Omega.7 gripper angle θ_Ω:

$$\alpha_g = \mathrm{clip}\left(\frac{\theta_{\max} - \theta_\Omega}{\theta_{\max} - \theta_{\min}}, 0, 1\right), \tag{7}$$

$$u_g = \min\big(\alpha_g G_{\mathrm{cue}} F_{\mathrm{cue,max}}, F_{\mathrm{cue,max}}\big), \qquad u_{h,z} = u_{h,z}^{\mathrm{base}} + u_g. \tag{8}$$

With $G_{\mathrm{cue}} = 0.3$ and $F_{\mathrm{cue,max}} = 1.0$ N, the cue was $u_g = 0.3\alpha_g$ N and therefore remained within $[0, 0.3]$ N. It was updated during every nominal 200-Hz supervisory cycle, with a more open Omega.7 gripper producing a larger cue. This signal

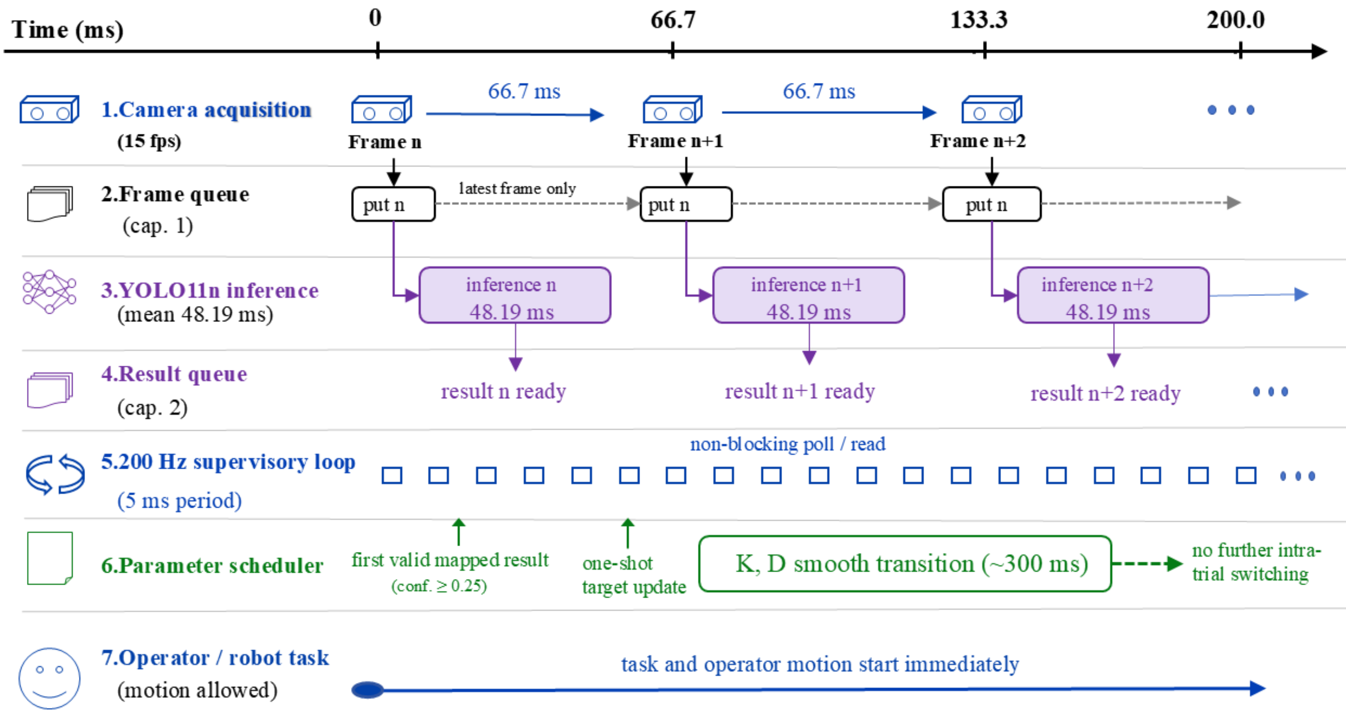

Figure 2: Schematic, not-to-scale timing of the asynchronous perception–control implementation. Color frames are acquired at 15 fps and passed through a single-slot frame queue to an independent YOLO11n process. Class labels and confidence scores are returned through a two-slot result queue, which is polled without blocking by the nominal 200-Hz supervisory loop. The first valid mapped result (confidence ≥ 0.25) triggers a one-shot target update. Translational stiffness, rotational stiffness, and the controller damping-scaling parameter are commanded to follow a smooth transition with a nominal duration of approximately 300 ms. Subsequent detections do not change the target within the same trial, while task execution and operator motion begin immediately. The schematic does not measure end-to-end trigger latency or trigger-to-contact timing.

Table 1: Runtime characteristics and data flow of the multi-rate mechatronic system.

Module	Timing or trigger	Data flow
Master input	200 Hz	Omega.7 translational position and buttons → master-state update.
Slave control	200 Hz	Desired pose and impedance parameters → Panda command.
Image acquisition	15 fps (66.7 ms/frame)	D435i color stream → 424×240 color image.
YOLO11n inference	Asynchronous; controlled-test performance reported in Section 4.1	Latest available image → class and confidence.
Strategy scheduling	First threshold-qualified detection	Class and confidence → target $\Theta(c)$.
Haptic rendering	200 Hz	Force-related state → Omega.7 force command.
Gripper command	Button event	Gripper input and command parameters → Franka Hand command.

Note: YOLO11n inference is executed in an independent process and does not block the supervisory loop.

represents only the Omega.7 gripper aperture; it contains no measurement of Franka Hand aperture, master–slave aperture error, grasp force, or button state.

For gripper execution, v_g was passed as the `speed` argument of the `grasp` API, whose remaining arguments were `width`, `force`, `epsilon_inner`, and `epsilon_outer`; it was also used by `move` during opening and non-grasp position adjustments. The strategy value F_g specified the requested `force` argument of `grasp()` in newtons; it was not an independently measured fingertip force. The target grasp width was 0 m. After contact and successful entry into the `HOLDING` state, the grasp command was not repeatedly retransmitted; the gripper maintained the grasp internally until `stop()` and the subsequent release or opening command.

Together, the three parameter channels act on distinct elements of the teleoperation system. Panda impedance defines the slave-side response to pose error and disturbance, the Omega.7 settings determine how the estimated external force is rendered to the operator, and the Franka Hand settings govern grasp execution. The proposed method assigns these settings in parallel from a single operational strategy. Coordination therefore occurs at the strategy layer through a consistent parameter bundle;

no dynamic coupling constraint or master–slave impedance-matching law is derived between channels. Haptic transparency and closed-loop fingertip-force accuracy were not evaluated.

2.3. Object-Conditioned Strategy-Level Multi-Subsystem Parameter Configuration

The framework uses a three-level mapping from detected object class to an operational strategy and then to a joint parameter vector,

$$\Theta(c) = \{K_t, K_r, \zeta, K_f, d, v_g, F_g\}. \tag{9}$$

The strategy classes represent combined grasping risks rather than material softness alone. The apple and banana were assigned to the fragility-priority strategy because lower contact impact, closing speed, and requested grasp force were preferred to limit squeezing and deformation, while their smooth surfaces still posed a risk of slip. The paper cup and bottle were assigned to the balanced strategy. Excessive force can deform a paper cup, whereas overly low settings may produce an insecure grasp; the bottle similarly requires a compromise between gentle handling and slip resistance. The mouse and scissors were assigned to the stability-priority strategy because their smooth or irregular surfaces and transfer demands—and, for the scissors, elongated geometry—placed greater emphasis on pose retention and grasp stability. These assignments were specific to the present objects, platform, and task; the associated parameter settings are listed in Table 2.

At run time, the first threshold-qualified result read by the supervisory loop triggered a one-time update of the parameters enabled in the active mode, after which the strategy remained locked for the rest of the trial. Translational stiffness, rotational stiffness, and the damping-scaling parameter followed a smoothstep interpolation over approximately $T_s = 0.30$ s. With $\tau = \mathrm{clip}((t - t_0)/T_s, 0, 1)$ and $s(\tau) = 3\tau^2 - 2\tau^3$, each interpolated quantity $q \in \{K_t, K_r, \zeta\}$ was updated as

$$q(t) = q_0 + s(\tau)(q_{\mathrm{target}} - q_0). \tag{10}$$

The smoothstep interpolation yields continuous trajectories for K_t, K_r, and ζ, with zero update rate at the start and end of the transition. It was introduced to reduce discontinuities in the commanded impedance parameters and is not interpreted as a formal transient-stability guarantee for the complete teleoperation system.

The controller recomputed $\mathbf{D}_{\mathrm{API}}$ from the current $\mathbf{K}$ and ζ during this transition. Enabled values of K_f, d, v_g, and F_g changed immediately at the strategy event. The update was intended to occur before first contact; however, because operator motion was not gated by perception and trigger–contact timestamps were not recorded, trial-level contact-preparatory timing could not be verified.

The class-to-strategy mapping is manually engineered. Its role is not to infer physical properties directly, but to convert a recognized object condition into a constrained, repeatable operating strategy spanning three subsystems. If an existing detector already provides a stable label for a new object, that label can be added to the mapping without retraining the controller. If the detector cannot identify or distinguish the object reliably, it must be retrained or replaced before a strategy can be assigned. The present system does not automatically infer fragility, friction, or other physical attributes for an unseen class.

2.4. Experimental Modes and Parameter Rationale

The fixed baseline vector was

$$\Theta_0 = (150,\ 10,\ 1.0,\ 0.5,\ 0.3,\ 0.10,\ 20), \tag{11}$$

with the same element order as $\Theta(c)$. Table 3 summarizes the five experimental conditions and the comparisons they support.

The visual display remained active throughout all formal trials in modes C–E.

The three strategies were represented by discrete operating points. Before formal data collection, two researchers qualitatively screened candidate operating points on representative objects. Settings associated with sustained oscillation, noticeable haptic discomfort, grasp failure, or visible object damage were excluded, and the three final operating points were fixed before the 135 formal trials. The fragility-priority point used 50 N/m translational stiffness, 8 N requested grasp force, and 0.02 m/s gripper speed to reduce contact and gripper-closing severity. The balanced point used 150 N/m, 15 N, and 0.05 m/s to avoid both excessive deformation and unreliable holding. The stability-priority point used 200 N/m, 20 N, and 0.10 m/s to favor pose retention and transfer stability. Table 4 summarizes the parameter effects, engineering constraints, and selected levels. These operating points were selected qualitatively for the present setup and should not be interpreted as material-specific damage limits.

2.5. Execution Procedure

Algorithm 1: Object-conditioned strategy-level parameter scheduling

1. Load the mode-specific initialization in Table 3 and start the nominal 200-Hz supervisory update; launch the visual process only in modes C–E.
2. Run object detection asynchronously. The supervisory thread polls the results without blocking and checks the confidence threshold.
3. If no detection meets the threshold, retain the initialized state. If the detected label has no explicit class-to-strategy entry, assign the balanced default strategy.
4. Use the first threshold-qualified strategy to update the parameter channels enabled by the current mode. Interpolate K_t, K_r, and the controller parameter ζ over approximately 300 ms and update the enabled haptic-interface and gripper settings immediately.
5. Lock the selected strategy for the remainder of the trial and continue teleoperation, haptic rendering, and gripper execution without further visual switching.
6. Record the active parameter state, task trajectory, duration, and outcome, and reset the system after the prescribed terminal procedure.

Table 2: Parameter settings for the three operational strategies. K_t is in N/m, K_r in N m/rad, d in N, v_g in m/s, and F_g is the requested `grasp()` force in N. ζ and K_f are dimensionless controller/API scaling parameters.

Strategy	Assigned objects	Slave impedance	Haptic interface	Gripper
Fragility-priority	Apple, banana	K_t = 50 N/m; K_r = 5 N m/rad; ζ = 0.8	K_f = 0.2; d = 0.3 N	v_g = 0.02 m/s; F_g = 8 N
Balanced	Paper cup, bottle	K_t = 150 N/m; K_r = 10 N m/rad; ζ = 1.0	K_f = 0.5; d = 0.4 N	v_g = 0.05 m/s; F_g = 15 N
Stability-priority	Mouse, scissors	K_t = 200 N/m; K_r = 13 N m/rad; ζ = 1.2	K_f = 0.7; d = 0.5 N	v_g = 0.10 m/s; F_g = 20 N

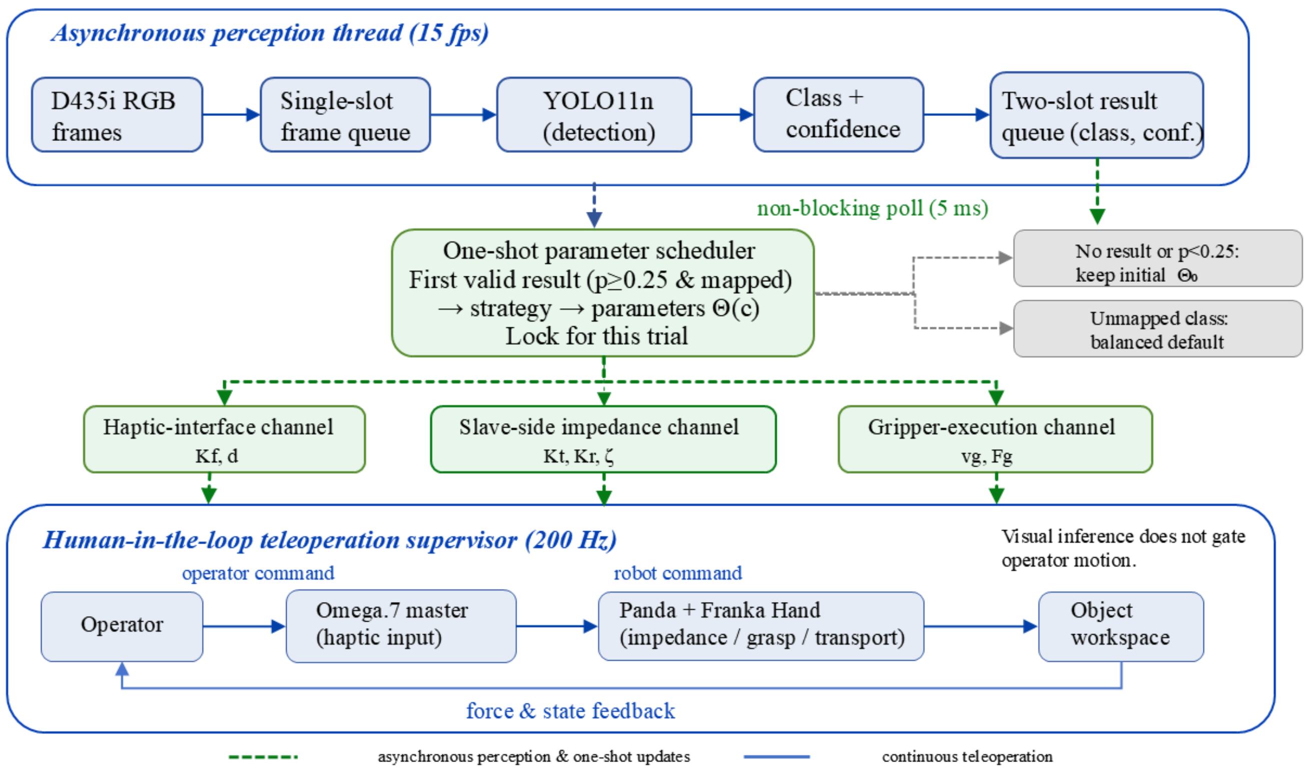

Figure 3: Object-conditioned strategy-level parameter-bundle architecture. D435i RGB frames are processed asynchronously by YOLO11n and returned as class–confidence results through bounded queues. The 200-Hz teleoperation supervisor polls the result queue without blocking. The first mapped result with confidence ≥ 0.25 is converted into one strategy event $\Theta(c)$, which atomically assigns the haptic-interface, slave-side impedance, and gripper-execution parameter groups and is locked for the trial. The diagram also shows the retained initial state for no or low-confidence results, the balanced default for an unmapped class, and the continuous operator–robot force/state-feedback loop. This is an implementation architecture; formal trials did not retain event-level trigger histories or synchronized trigger–contact timestamps.

2.6. Bounded Fallback and Safety Measures

A misclassification as another known class invoked the strategy assigned to that class. A detected label without an explicit mapping invoked the balanced default strategy, whereas a missing or low-confidence detection caused no update. Every fallback outcome was confined to one of the three predefined parameter sets, although a bounded parameter set may still be inappropriate for a misclassified object. Per-trial visual events were outside the formal logging scope, so misclassifications and default-branch activations were not individually recorded.

In addition to these software fallbacks, robot collision detection, Franka Hand force settings, zero-force commands on exit, a standardized initial pose, and a manual emergency stop provided operational safeguards. The Omega.7 was operated using its standard driver and hardware configuration. The implemented mapping limited the additional master-gripper-aperture cue to 0.3 N. The baseline force-feedback command was transmitted without an additional software saturation layer, and saturation status was not logged.

3. Experimental Design

3.1. Evaluation Objectives and Analysis Plan

The evaluation focused primarily on task completion time under joint parameter-bundle configuration relative to the planned baselines. Success rate and Raw NASA-TLX were reported as

Table 3: Experimental conditions, channel-update logic, and comparison roles for the five modes.

Mode	Visual cue / selection	Slave impedance	Haptic interface	Gripper execution	Comparison role
A: Fixed-parameter baseline	None	—	—	—	Fixed-parameter baseline.
B: Manual-selection workflow baseline	No visual cue; operator selects a complete strategy after timing starts.	Manual	Manual	Manual	Workflow baseline; timing includes manual identification and selection.
C: Joint parameter-bundle configuration	Class, strategy, confidence, and colored bounding box	Auto	Auto	Auto	Proposed architecture; C–D compares the full bundle with the displayed fixed-parameter condition, and C–E compares the full bundle with impedance-only reconfiguration.
D: Visual-semantic display with fixed parameters	Same as C	—	—	—	Isolates the visual-semantic display with fixed parameters.
E: Impedance-only reconfiguration	Same as C	Auto	—	—	Isolates impedance-only reconfiguration.

Note: Auto denotes a one-shot update after the first valid visual detection; Manual denotes manual strategy selection after timing starts; — retains the initialized setting. In modes C–E, no detection or confidence below 0.25 retains the initialized state. A detected class without an explicit mapping invokes the balanced strategy; in mode D, the visual-semantic output is displayed while Θ_0 is retained. Vision is not used in modes A and B. The C–E contrast changes the haptic-interface and gripper parameter groups together and therefore does not identify their individual effects or an interaction between them.

Table 4: Parameter design space, engineering constraints, and selected levels.

Parameter	Low-to-high operational effect	Engineering constraint	Selected levels
Slave impedance: K_t	Greater translational compliance and lower impact → greater positional stability.	No sustained oscillation during qualitative screening.	50 / 150 / 200 N/m
Slave impedance: K_r	Greater rotational compliance about the fixed desired orientation → greater resistance to rotational disturbance.	No sustained oscillation during qualitative screening.	5 / 10 / 13 N m/rad
Slave impedance: ζ	Lower → higher controller damping scaling.	Avoid sustained oscillation and excessive sluggishness.	0.8 / 1.0 / 1.2
Haptic interface: K_f	Weaker → stronger rendered external-force cue.	Acceptable Omega.7 response.	0.2 / 0.5 / 0.7
Haptic interface: d	Greater sensitivity to small forces → greater suppression of small signals.	Haptic-interface jitter and noise.	0.3 / 0.4 / 0.5 N
Gripper: v_g	Lower-impact motion → faster grasp establishment.	Franka Hand execution range.	0.02 / 0.05 / 0.10 m/s
Gripper: F_g	Lower → higher requested grasp force.	Franka Hand force setting.	8 / 15 / 20 N

secondary descriptive evidence of task success and subjective workload. Master-side trajectory length and pause count were included as exploratory process measures. Mode A served as the fixed-parameter baseline; mode B was a manual-selection workflow baseline, in which the operator identified the object and selected a strategy after timing began; and mode D was the visual-semantic display with fixed parameters. Mode E implemented impedance-only reconfiguration, retaining visual-semantic mapping and slave-side impedance updates but not the haptic-interface or gripper settings. The primary C–E comparison therefore evaluated the complete parameter bundle at the system level against impedance-only reconfiguration; it was not designed to separate the independent contributions of the added parameter groups or to test an interaction between them. All statistical comparisons are limited to repeated measurements in the present three-operator sample.

3.2. Operators and Test Objects

Three operators completed the main experiment (P01–P03; 23–24 years old; two male and one female; all right-handed). The Omega.7 was positioned to the operator's left and was operated with the left hand in every trial. All operators had received basic teleoperation training and completed a 10–15 min familiarization session immediately before formal data collection to practice master-device motion, gripper input, and the task procedure. All operators provided written informed consent.

The six objects were selected to represent different contact, grasping, and transfer requirements and were assigned to the fragility-priority, balanced, or stability-priority strategy (Table 5). These assignments were used solely to determine the parameter settings in this experiment and should not be interpreted as a general classification of material properties.

3.3. Experimental Modes and Trial Structure

The five modes and their comparison logic are defined in Table 3. Modes C, D, and E used the same visual-semantic display. The C–D contrast therefore compares joint parameter-bundle configuration with the visual-semantic display with fixed parameters, whereas C–E compares the full bundle with impedance-only reconfiguration while holding the designed interface constant. The D–A contrast assesses the complete visual-semantic display—class, mapped strategy, confidence, and bounding box—and does not isolate the class label. Mode B required manual

Table 5: Test objects, assigned strategies, and principal task risks.

Object	Assigned strategy	Physical profile	Principal task risk
Apple	Fragility-priority	~200 g; smooth; ∅70–80 mm	Impact or slip; gentle contact required.
Banana	Fragility-priority	~120 g; smooth; 20 × 180 mm	Compression-induced deformation and slip.
Paper cup	Balanced	~5 g; paper; ∅75×90 mm	Easily deformed; stable grasp required.
Bottle	Balanced	~30 g; smooth plastic; ∅65×200 mm	Slip; balance between efficiency and stability.
Mouse	Stability-priority	~100 g; smooth plastic; 65 × 120 × 35 mm	Rigid irregular surface; transfer slip.
Scissors	Stability-priority	~150 g; metal and plastic; 50 × 170 × 15 mm	Rigid elongated geometry; high orientation demand.

Table 6: Allocation of matched task blocks and trials by strategy and object.

Object	Blocks	Trials
Fragility-priority		
Apple	4	20
Banana	5	25
Balanced		
Paper cup	5	25
Bottle	4	20
Stability-priority		
Mouse	5	25
Scissors	4	20
Total (six objects)	27	135

Note: Each block contains one trial in each of the five modes.

object identification and button-based strategy selection after timing had started but before robot motion could begin. Its comparison with mode C therefore includes both automation of the selection workflow and differences in visual-interface presentation; the separate decision time was not logged and this contrast is not interpreted as an isolated controller effect.

The experiment comprised 27 matched task blocks. Each block was defined by a fixed operator, object, and repetition index and contained one trial in each of modes A–E, yielding $27 \times 5 = 135$ trials. Within each operator and strategy category, the two objects were assigned to the three repetitions in a fixed 2:1 ratio. This assignment was held constant across the five modes for a given operator and strategy, preserving the matched comparison, while the 2:1 allocation was reversed across operators. The resulting object counts yielded the approximate 5:4 balance within each strategy shown in Table 6.

The five mode orders were preassigned at the operator-by-strategy group level using nine different sequences, so no single fixed order was used. The schedule distributed modes across early and late positions but did not constitute a complete Latin-square design. Across the nine sequences, mode C occurred in position four in three groups, whereas mode E occurred in position five in three groups. The exact sequences are reported in Supplementary Table S2. With only three operators, order, object, and operator effects could not be fully separated; the C–E difference is therefore interpreted as a within-sample paired result rather than an order-free population effect.

3.4. Task and Procedure

Before each trial, the test object was manually placed within a predefined work region and at an approximately prescribed orientation. Small reset errors in position and orientation were therefore unavoidable, but the same placement protocol was used in all modes. Only translational master motion was mapped; the desired Panda end-effector orientation remained fixed throughout the trial.

Each trial comprised reset, approach, grasp, transfer, release, and termination (Fig. 3). In modes C–E, camera acquisition, visual inference, and teleoperation began concurrently at task initiation; visual output did not gate operator motion. In mode B, timing began before object identification and strategy selection, and robot motion was permitted only after the operator completed the button selection. No separate timestamp marked the end of this selection step.

A trial was classified as successful when grasping, transfer, and placement were completed without a drop, observable slip, or visible damage. Complete separation of the object from the gripper was classified as a drop; visible relative motion between object and gripper during grasp or transfer was classified as slip; and a clear indentation, crushing deformation, or other visible damage was classified as damage. A designated researcher applied these predefined criteria in real time. Trials were not systematically video-recorded, and the labels were not independently or blindly re-evaluated.

To maintain a common timing endpoint, failure did not stop the timer immediately. Regrasping after a drop was not permitted; the failure was recorded, and the operator completed the prescribed return-and-termination procedure until the robot and gripper reached the predefined terminal state. The same endpoint was therefore applied to successful and failed trials. The system recorded the master-side trajectory, gripper input, active control parameters, task duration, and outcome. It did not retain an event-level history of visual classes, confidences, strategy-trigger timestamps, or synchronized trigger–contact timestamps. Consequently, the retained active-state record cannot retrospectively establish whether the intended strategy was selected at a verified pre-contact time in each trial.

3.5. Evaluation Measures

Task completion time was the primary endpoint. Success rate was reported descriptively, and master-side trajectory length and pause count were used as exploratory process measures.

Subjective workload was assessed with Raw NASA-TLX [26]. Each of the six dimensions was rated from 0 to 100, and their unweighted arithmetic mean was used as the Raw NASA-TLX score.

Each operator completed one questionnaire immediately after the three trials in a given strategy × mode condition. These trials combined the two objects assigned to that strategy in the operator-specific 2:1 ratio. The workload dataset therefore contained 3 operators ×3 strategies ×5 modes = 45 questionnaires, or nine questionnaire units per mode. The resulting scores represent strategy-level workload under each mode and do not support object-specific workload comparisons or population-level workload inference.

Visual performance was characterized in a separate controlled image test using class accuracy, class-to-strategy mapping accuracy, confidence, and per-image processing time. A pause was defined as a continuous interval during which master-device speed remained below 0.005 m/s for at least 0.30 s. Pauses were detected by differentiating the raw master-side trajectories, which were sampled at approximately 200 Hz.

3.6. Statistical Analysis

Completion time across the five modes was first compared using a Friedman test, with the task block—defined by the same operator, object, and repetition index—as the pairing unit. When the omnibus test was significant, paired Wilcoxon signed-rank tests were performed with Holm–Bonferroni correction for multiple comparisons. For the primary C–E comparison, we report the paired mean difference, relative change, and a 95% bootstrap confidence interval based on 10,000 resamples of the matched task blocks. We also examined effect direction for each operator and performed a leave-one-operator-out analysis to assess whether a single operator dominated the overall difference. The paired C–E success outcomes were tabulated, and an exact McNemar test was reported as a supplementary analysis because few discordant pairs were observed. Raw NASA-TLX scores were paired descriptively across nine operator × strategy units per mode; each score summarized three mixed-object trials, and no bootstrap interval was calculated. Trajectory length and pause count were treated as exploratory measures. Because the 27 task blocks were nested within only three operators, the inferential tests characterize repeated-measures differences within the present sample and do not support population-level inference across operators. Results are reported as both median [IQR] and mean ± SD. Figures 4–9 were generated from frozen data using verified Python/Matplotlib scripts.

4. Results

4.1. Controlled Visual Test and Asynchronous Integration

The separate visual test set contained 30 photographs per class, for a total of 180 images. The images were acquired under fixed viewpoint, background, and lighting conditions and were not used for model training or parameter selection. The model correctly classified all 180 images and mapped every prediction to the intended strategy. Mean confidence was 0.853, and mean per-image inference wall-clock time was 48.19 ms (Table 7 and Figs. 4 and 5). These results characterize the six known object instances under controlled imaging conditions. They do not quantify online trigger correctness, false-trigger/default-branch frequency, end-to-end trigger latency, or completion of configuration before contact during the 135 teleoperation trials.

Table 7: Class-wise results of the closed-set controlled visual test.

Object	Images	Mean confidence	Mean time (ms)
Apple	30	0.771	49.89
Banana	30	0.948	48.02
Bottle	30	0.726	48.57
Paper cup	30	0.820	46.62
Mouse	30	0.914	46.79
Scissors	30	0.938	49.27

Note: All 180 images were classified correctly and mapped to the intended strategy under the fixed viewpoint, background, and lighting conditions. This table does not quantify online trigger correctness, false triggers, or trigger-to-contact timing during teleoperation.

Figure 4: Confusion matrix for the separate closed-set controlled visual test (180 images; 30 images per class). Each cell gives the number of images for the corresponding true and predicted class. All observations lie on the diagonal under the fixed viewpoint, background, and lighting conditions.

In the supervisory-loop implementation, the independent YOLO process, bounded queues, and non-blocking polling kept visual inference outside the synchronous call path of the nominal 200-Hz update. The median logged cycle time was 5.07 ms.

4.2. Results Across the Five Modes

Mode C had the lowest median completion time, the highest observed success rate, and the lowest Raw NASA-TLX score (Table 8 and Figs. 6 and 7). It also had the lowest median

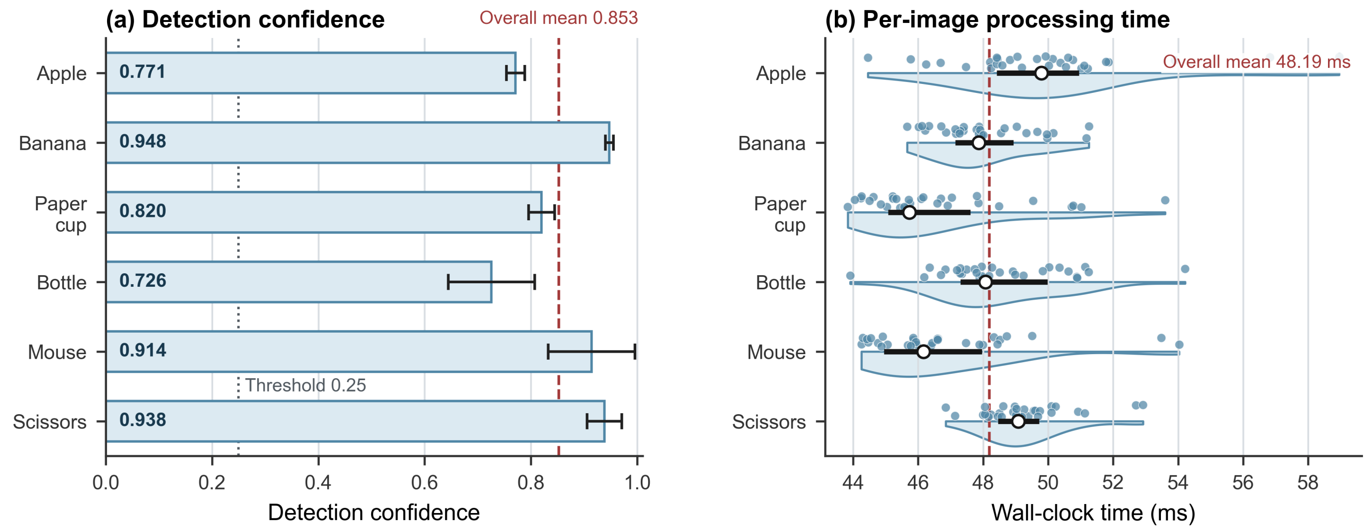

Figure 5: Detection confidence and per-image inference wall-clock time for the closed-set controlled 180-image visual test (n = 30 images per class). (a) Class-wise confidence: horizontal bars show mean ± SD; the dotted and dashed vertical lines mark the decision threshold (0.25) and overall mean (0.853). (b) Per-image processing time: half violins show class-wise distributions, dots show individual images, horizontal segments and open circles show the IQR and median, and the dashed vertical line marks the overall mean (48.19 ms). These timing data are not end-to-end task-trigger latency measurements.

Table 8: Descriptive results across the five experimental modes.

Mode	Completion time (s)	Trajectory length (m)	Success rate	Raw NASA-TLX
A	21.18 [20.62, 22.08]	0.757 [0.693, 0.816]	22/27 (81.5%)	62.50 [59.67, 64.50]
B	20.89 [20.12, 21.83]	0.787 [0.721, 0.861]	21/27 (77.8%)	56.17 [55.00, 59.33]
C	**19.57 [18.41, 20.05]**	**0.697 [0.660, 0.769]**	**26/27 (96.3%)**	**48.67 [47.67, 51.83]**
D	20.79 [20.32, 21.16]	0.722 [0.678, 0.768]	24/27 (88.9%)	60.33 [57.33, 62.50]
E	20.73 [19.95, 22.25]	0.732 [0.678, 0.799]	24/27 (88.9%)	53.67 [51.83, 57.83]

Note: Values are median [IQR], except success rate. Completion time, trajectory length, and success rate use n = 27 trials per mode; Raw NASA-TLX uses n = 9 strategy-level questionnaire units per mode. Mode definitions are given in Table 3; trial-level performance data are provided in Supplementary Table S1.

master-side trajectory length, an exploratory measure. Using the trial-level values in Supplementary Table S1, mean completion time in mode C (19.28 ± 1.30 s) was lower than in modes A (21.42 ± 1.58 s), B (21.01 ± 1.61 s), D (20.91 ± 1.10 s), and E (21.07 ± 1.56 s) by 10.0%, 8.2%, 7.8%, and 8.5%, respectively. Subsequent inferential analyses focused on completion time as the primary endpoint.

Consistent with the descriptive results, the Friedman test across the 27 matched task blocks indicated differences in completion time among the five modes ($\chi^2(4) = 30.904$, $p < 0.001$). Holm-adjusted paired Wilcoxon tests showed lower completion times in mode C than in modes A, B, D, and E (all $p < 0.01$). Because the task blocks were nested within only three operators, these statistics describe repeated-measures differences within the present sample and do not support population-level inference across operators. Descriptively, mode C had the lowest Raw NASA-TLX score in all nine operator × strategy units; given the three-operator sample, no population-level workload inference was made.

4.3. Primary Comparison: Complete Parameter Bundle Versus Impedance-Only Reconfiguration

Table 9 summarizes the primary C–E comparison.

The median completion times were 19.57 s in mode C and 20.73 s in mode E (Fig. 8). With $\Delta T = T_E - T_C$, the mean paired difference was 1.79 s and the bootstrap 95% CI was [1.10, 2.51] s, corresponding to a mean reduction of 8.5%. Completion time was lower in mode C in 22 of the 27 matched blocks. The mean differences for P01, P02, and P03 were 1.66, 2.56, and 1.16 s, respectively. Across the six objects, the relative differences ranged from 3.3% to 13.2% (Fig. 9).

As a post hoc sensitivity analysis, the C–E completion-time comparison was repeated after retaining only the 23 matched blocks in which both trials were successful. The mean paired difference remained 1.80 s, and 19 of the 23 blocks favored mode C. This descriptive sensitivity result indicates that the primary mean difference was not driven by the four blocks with a failure in either mode.

For paired success outcomes, 23 blocks succeeded in both modes, three succeeded in mode C and failed in mode E, one failed in mode C and succeeded in mode E, and none failed in both modes. With only four discordant pairs, the exact two-

Table 9: Within-sample primary comparison of the complete parameter bundle and impedance-only reconfiguration.

Measure	C, median [IQR]	E, median [IQR]	Mean Δ (E–C), 95% CI	Operator direction
Completion time (s)	19.57 [18.41, 20.05]	20.73 [19.95, 22.25]	1.79 [1.10, 2.51]	3/3 favored C
Trajectory length (m)	0.697 [0.660, 0.769]	0.732 [0.678, 0.799]	0.024 [-0.014, 0.059]	Mixed
Raw NASA-TLX	48.67 [47.67, 51.83]	53.67 [51.83, 57.83]	4.87 [not calculated]	3/3 favored C

Note: Positive Δ values favor mode C. Completion time and trajectory length use n = 27 matched task blocks; Raw NASA-TLX uses n = 9 questionnaire units per mode. Intervals are matched-task-block bootstrap 95% CIs from 10,000 resamples and quantify uncertainty across the observed blocks, not a population-level participant effect; no bootstrap interval was calculated for Raw NASA-TLX.

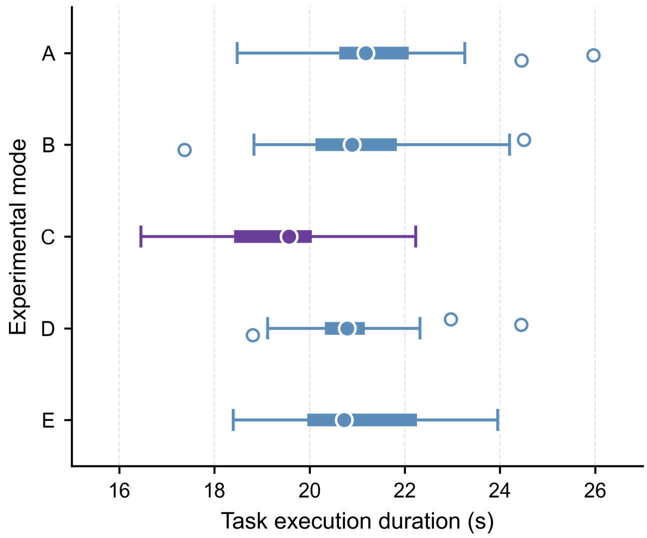

Figure 6: Task execution duration across the five experimental modes: fixed-parameter baseline (A), manual-selection workflow baseline (B), joint parameter-bundle configuration (C), visual-semantic display with fixed parameters (D), and impedance-only reconfiguration (E). Each mode contains 27 matched task blocks. Filled circles show medians, thick horizontal segments show IQRs, thin horizontal segments with end caps show 1.5 × IQR Tukey whiskers, and small open circles show observations beyond the whiskers. This descriptive distribution plot does not display the block pairing; the primary C–E pairing is shown in Fig. 8.

sided McNemar test did not establish a difference (p = 0.625). The observed success-rate difference is therefore interpreted as supplementary descriptive evidence rather than as a confirmed mode effect.

The median master-side trajectory lengths were 0.697 m in mode C and 0.732 m in mode E. The mean paired difference was 0.024 m, with a bootstrap 95% CI of [-0.014, 0.059] m. Pause counts were lower on average in mode C but had the same median in modes C and E.

4.4. Exploratory Process Measure: C–E Pause Analysis

Pause count was analyzed as an exploratory process measure using master-side trajectories sampled at approximately 200 Hz. A pause was defined as a continuous interval in which master-device speed remained below 0.005 m/s for at least 0.30 s, excluding momentary decelerations and capturing more sustained interruptions. Mode C had a median pause count of 3 [2, 3.5] and a mean of 2.74 ± 1.23, whereas mode E had a median of 3 and a mean of 3.41 ± 1.67. Although mode C had a lower mean in each strategy class, the two modes had the same overall median. No formal hypothesis test was prespecified or performed for pause count, and no threshold-sensitivity analysis was available.

4.5. Failure Analysis

Eighteen failures occurred across the 135 trials: 5, 6, 1, 3, and 3 in modes A–E, respectively. The failures included object drops, observable slip, and visible damage, as summarized in Table 10. One designated researcher recorded outcomes in real time using the predefined criteria in Section 3.4; no systematic video recording or independent re-rating was available.

Mode C had one failure in 27 trials, the lowest observed count among the five modes under the present experimental conditions.

4.6. Consistency Across Operators and Objects

Figure 9 reports paired C–E differences by operator and object. At the operator level, the mean paired differences $\Delta T = T_E - T_C$ for P01, P02, and P03 were 1.66, 2.56, and 1.16 s, respectively. The corresponding median [IQR] differences were 1.91 [0.60, 3.14], 2.63 [2.19, 3.34], and 1.60 [-0.53, 1.73] s. Mode C was faster in 7/9, 9/9, and 6/9 task blocks, respectively. After P01, P02, and P03 were excluded in turn, the mean differences in the remaining blocks were 1.86, 1.41, and 2.11 s, respectively; all continued to favor mode C.

At the object level, mode C also had the shortest mean completion time among the five modes for each of the six objects (Table 11):

Relative to mode E, the mean completion-time reductions in mode C were 3.3% for the bottle, 5.5% for the banana, 8.1% for the apple, 8.7% for the mouse, 11.7% for the paper cup, and 13.2% for the scissors. The direction was consistent across all objects, although the magnitude varied.

5. Discussion

5.1. Interpreting the Five-Mode Evaluation

Visual-semantic display and automatic configuration play distinct experimental roles. The five modes differed in how they used object information and which parameter groups they reconfigured. Mode A used the fixed parameter vector without a visual interface. Mode B also omitted the visual interface; after timing began, the operator identified the object,

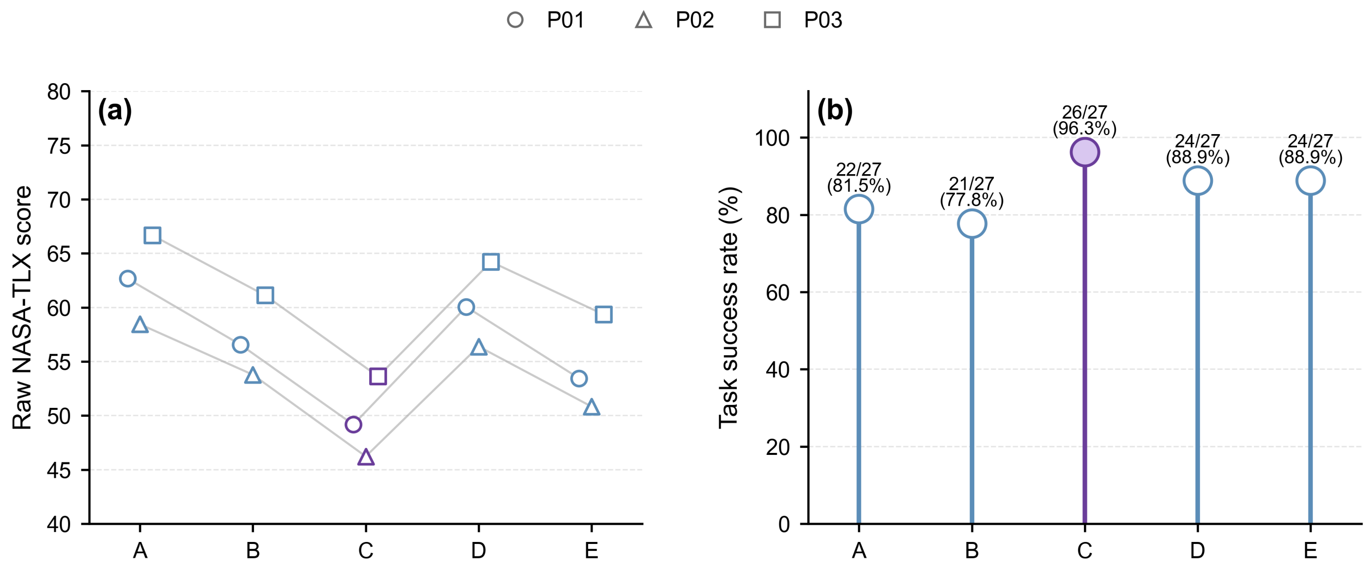

Figure 7: Human workload and observed task success across the five experimental modes. (a) Raw NASA-TLX scores: connected markers show each operator's mean across three strategy-level questionnaire units per mode (n = 9 questionnaire units per mode); marker shape identifies the operator. (b) Observed successful trials out of 27 for each mode. These panels are descriptive and do not display confidence intervals or inferential tests.

Table 10: Observed failures and representative observations by experimental mode.

Mode	Failures/total	Representative observations
A	5/27	Paper-cup deformation and unstable positioning of the scissors.
B	6/27	The manual-selection workflow included additional judgment and switching steps; the available records do not support attribution of specific failure causes.
C	**1/27**	Mouse slip during transfer.
D	3/27	Object drops or loss of grasp during holding.
E	3/27	Observed instability during grasp establishment or transfer.

Note: Mode definitions are given in Table 3. Observations are descriptive and were recorded in real time.

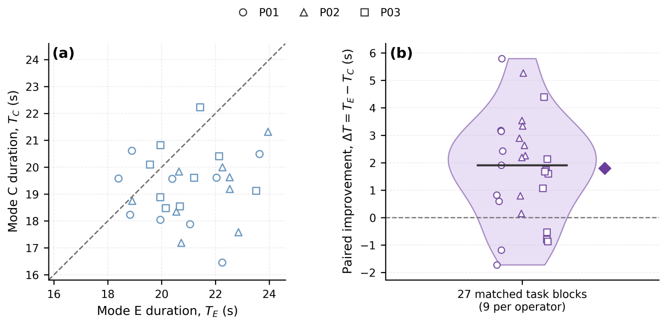

Figure 8: Paired completion-time comparison between modes C and E for 27 matched task blocks across three operators. (a) Each point represents one matched block; points below the identity line have a shorter duration in mode C. (b) Distribution of the paired difference $\Delta T = T_E - T_C$; positive values favor mode C. The violin, horizontal segment, and diamond show the distribution, median, and mean, respectively. The observed mean difference was 1.79 s (matched-block bootstrap 95% CI, 1.10–2.51 s), and 22/27 blocks favored C; the interval is conditional on the observed blocks.

Table 11: Mean completion time by object and experimental mode.

Object	A	B	C	D	E
Apple	20.46	21.40	**19.25**	20.81	20.94
Banana	20.07	20.85	**19.49**	21.06	20.64
Paper cup	22.36	20.27	**18.69**	20.38	21.17
Bottle	22.11	22.14	**19.63**	20.78	20.30
Mouse	21.74	20.93	**19.75**	20.74	21.64
Scissors	21.79	20.70	**18.84**	21.83	21.70

Note: Values are mean completion time in seconds. Mode definitions are given in Table 3; bold indicates the lowest mean in each row.

selected a strategy, and only then initiated robot motion. The B–C difference therefore includes removal of a manual decision-and-selection step and does not isolate controller performance. In modes C–E, the visual-semantic display remained active throughout the formal trials. Mode D displayed the same class, mapped strategy, confidence, and bounding box as mode C but retained fixed parameters. The C–D contrast is therefore the clearest comparison of the complete parameter bundle with the visual-semantic display held constant.

Mode C was faster and yielded lower Raw NASA-TLX

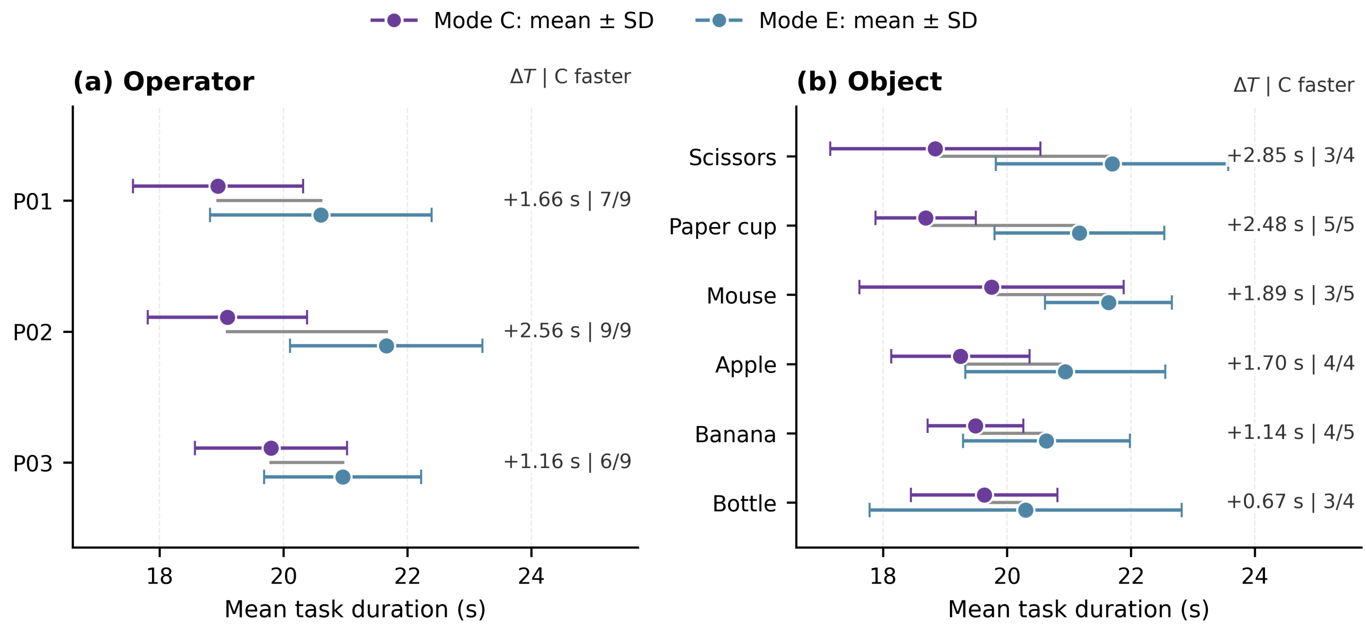

Figure 9: Descriptive mean task duration for modes C and E, stratified by operator and object. Purple and blue points show mean ± SD for modes C and E, respectively; gray connectors link the paired means. Labels report $\Delta T = T_E - T_C$ and the number of matched blocks favoring mode C (nine per operator and four or five per object). The object-level summaries are descriptive because each contains only four or five blocks.

scores than mode D while also reconfiguring the full parameter vector. Because modes C and D used the same visual-semantic display, the observed pattern is consistent with an additional task-level benefit from automatic full-parameter reconfiguration beyond the display itself. By contrast, the D–A comparison represents the effect of the complete displayed cue rather than that of the class label alone.

Overall, mode C had the shortest completion time and the lowest Raw NASA-TLX score in the present sample. The C–D and C–E comparisons are consistent with a within-sample task-level advantage of automatic parameter-bundle configuration beyond either the default visual-semantic display or impedance-only reconfiguration. These conclusions apply only to the designed default interfaces, the six known object instances, and the present repeated-measures sample. The C–E comparison provides the primary system-level evidence for the complete parameter bundle.

5.2. System-Level Effect of the Complete Parameter Bundle

Impedance-only reconfiguration did not match the observed performance of the full parameter bundle. The C–E comparison is the primary system-level contrast in this study. Both modes used the same class-to-strategy mapping, impedance settings, and visual-semantic display. Mode C additionally updated the master-side haptic gain, force dead zone, gripper speed, and requested grasp-force setting. Because the comparison removes the haptic-interface and gripper parameter groups together, it assesses the overall difference between the complete parameter bundle and impedance-only reconfiguration but cannot identify the contribution of each added channel or test an interaction between groups.

Empirically, the paired mean E–C completion-time difference was 1.79 s, with a matched-block bootstrap 95% CI of [1.10, 2.51] s. The confidence interval for master-side trajectory length crossed zero, indicating no clear reduction in geometric path length. Mode C also had a lower mean pause count, but the records do not localize this difference to grasp, transfer, or release and contain no separate measure of corrective action.

The shorter completion time observed in mode C was therefore associated with the added haptic-interface and gripper settings as a combined bundle. Impedance settings determine the robot response to pose error and disturbance, master-side settings alter how estimated external force is rendered, and gripper settings govern the speed and requested force used to establish the grasp. The experiment evaluates the latter two parameter groups together; it neither identifies their separate contributions nor demonstrates that the commanded Franka Hand force equals the actual fingertip contact force. Channel-specific ablations are required before the observed difference can be attributed to either group separately.

The trajectory-length interval crossed zero. Thus, the completion-time advantage of mode C cannot be attributed simply to a shorter geometric path. The pause-count result remains exploratory and does not establish fewer interruptions, greater movement continuity, or any causal mechanism, particularly because no separate correction or regrasp measure was available.

Because the configuration was evaluated as a bundle, the failure results likewise cannot be attributed to any single parameter channel.

5.3. *Asynchronous Perception–Control Integration*

A one-shot update does not require perception to operate at the supervisory-loop rate. Image acquisition at 15 fps and independent inference exchanged data with the nominal 200-Hz update through bounded queues. The single-slot frame queue limited stale images, while non-blocking polling prevented the supervisory loop from waiting for visual output. This architecture is appropriate when perception supports a single configuration event after task initiation rather than continuous visual servoing.

The implementation therefore demonstrates asynchronous, non-blocking integration in the software architecture, but it does not establish hard real-time performance, measured end-to-end trigger latency, or completed pre-contact configuration in individual trials.

5.4. *Variation Across Operators and Objects*

At the operator level, the mean paired difference was positive for all three operators and remained positive when each operator was excluded in turn. Nevertheless, the participant sample was limited to three individuals.

At the object level, mode C was faster than mode E for all six objects, with mean reductions ranging from 3.3% to 13.2%. The differences were smaller for the bottle and banana and larger for the paper cup and scissors. This variation may reflect differences in grasp pose, deformation risk, and orientation-stability requirements. However, because only four or five matched task blocks were available for each object, the direction and magnitude are reported descriptively without assigning a specific mechanism to the object-level differences.

5.5. *Relation to Previous Work and Mechatronic Design Implications*

Previous tele-impedance and variable-impedance approaches commonly adapt remote impedance according to human state, contact force, demonstrations, or task phase [7–13, 27]. Visual methods have translated object attributes, geometry, and vision–language inputs into stiffness or impedance matrices [17, 18, 20, 21], while automatic switching, shared control, and multi-stiffness interfaces have addressed mode selection and control-authority allocation [19, 22, 23, 28, 29]. Teleoperated grasping has also used tactile deformation and grip-safety information to regulate gripper impedance [24]. Relative to these approaches, the present contribution is a constrained systems architecture in which one semantic strategy event atomically assigns a consistent operating state across slave mechanical response, operator-facing haptic rendering, and gripper execution. This coordination is semantic and task-level rather than based on a dynamically derived coupling law. Bounded queues decouple low-rate perception from the nominal high-rate supervisory update, while non-blocking polling, one-shot triggering, and strategy locking keep configuration transactional and prevent repeated intra-trial changes. The five-mode experiment then compares joint parameter-bundle configuration with the fixed-parameter baseline, manual-selection workflow baseline, visual-semantic display with fixed parameters, and impedance-only reconfiguration.

From a design perspective, this principle may be useful in flexible sorting, remote maintenance, and unstructured dismantling, where a limited set of recurring object or task classes can be associated with validated operating strategies. Scaling beyond a small class set would require systematic maintenance of the mapping and potentially more than three strategy levels. Additional sensing, such as tactile or contact-state information, could support post-contact verification or correction without placing the visual process in the high-rate supervisory path.

5.6. *Limitations and Future Work*

A timing-verification limitation is that the formal trial logs did not retain complete visual-event histories or synchronized strategy-event–contact timestamps. The trigger–contact interval could therefore not be reconstructed retrospectively for individual trials. The framework should therefore be interpreted as designed for contact-preparatory configuration rather than as having verified pre-contact timing in every trial. This limitation does not invalidate the observed C–E completion-time difference, but it limits attribution of that difference to a verified pre-contact, object-conditioned configuration event. The 180-image test characterizes recognition under controlled imaging conditions but does not provide trial-level evidence of trigger correctness, fallback use, or completion before contact during the 135 teleoperation trials.

No formal stability or passivity analysis was conducted for the time-varying human-in-the-loop multi-channel system, including the immediate haptic-interface and gripper updates. The qualitative operating-point screening and smooth impedance transition should therefore not be interpreted as a closed-loop transient-stability guarantee.

The study involved only three operators, with repeated measurements from the same participants. The inferential tests therefore describe within-sample repeated-measures differences; a larger participant study is required to assess population-level generalizability. All operators were right-handed but operated the left-positioned Omega.7 with the left hand. Only translational master motion was mapped, and the end-effector orientation remained fixed. Objects were manually reset within a prescribed region and approximate orientation, while the nine preassigned mode sequences did not provide complete positional balance. The results therefore apply to the present device layout, training conditions, motion mapping, object-placement protocol, and participant sample.

The class-to-strategy mapping is also manual. A new label can be added without controller retraining only if the detector already recognizes it reliably; otherwise, the detector must be replaced or retrained. The system does not automatically infer unseen physical properties or an appropriate strategy. The three strategy levels are practical abstractions for the present task and may become insufficient as the number and diversity of objects increase.

The parameter sets were operating points selected through qualitative screening rather than optimization or material-strength testing. Because the C–E comparison removes the haptic-interface and gripper parameter groups together, it cannot separate their contributions. Future ablations should independently remove haptic-gain, dead-zone, and gripper scheduling. The baseline Omega.7 force-feedback command had no additional software saturation, and saturation status was not logged. Future implementations should add an explicit software force limit and record both commanded and limited outputs. The Franka Hand `force` value specified a requested grasp setting rather than a measured fingertip force; direct fingertip-force measurements are needed to validate its physical effect.

Success and failure were judged in real time by one designated researcher using predefined criteria. Trials were not systematically video-recorded or independently re-rated, so slip and visible-damage labels retain an observational component. Failure did not end timing immediately, and a dropped object could not be regrasped; instead, the operator completed the common terminal procedure. This protocol avoids artificially short failure times but does not provide stage-specific failure durations. The records also lack a separate selection time for mode B, quantitative contact-quality measures, and a direct measure of corrective action.

Pause count used an engineering threshold of 0.005 m/s sustained for at least 0.30 s. No threshold-sensitivity analysis or prespecified inferential test was performed, so this result remains exploratory. Visual inference was kept outside the synchronous call path of the nominal 200-Hz update; hard real-time performance was not evaluated. Future work should add synchronized perception, contact, interface, and mode-selection logs; perform pause-threshold sensitivity analysis; record video for independent outcome assessment; incorporate active orientation control and less constrained object placement; and recruit a larger, more diverse participant sample.

6. Conclusion

This study presented an object-conditioned strategy-level parameter-bundle configuration framework for haptic teleoperation. A semantic mapping assigned each detected object class to an operational strategy that jointly specified slave-side impedance, master-side haptic rendering, and gripper execution. An asynchronous one-shot update was designed to integrate low-rate visual perception without placing inference in the nominal 200-Hz supervisory call path.

Across 27 matched task blocks comprising 135 physical trials, the complete parameter bundle reduced mean completion time by 1.79 s (8.5%; matched-block bootstrap 95% CI, 1.10–2.51 s) relative to impedance-only reconfiguration. Completion time favored the complete bundle in 22 of 27 matched blocks and in the mean for each operator and object, while success rate and Raw NASA-TLX were descriptively best in this mode. The post hoc sensitivity analysis restricted to matched blocks in which both trials succeeded yielded a similar mean difference of 1.80 s. No clear reduction in master-side trajectory length was observed.

These results describe a within-sample completion-time advantage of the complete parameter bundle under the tested conditions. Because the haptic-interface and gripper parameter groups were removed together in the primary comparison, their individual causal contributions remain unresolved, and the study does not demonstrate a dynamic coupling or synergy mechanism. Formal trials did not retain event-level visual histories or synchronized trigger–contact timestamps. Thus, they do not verify online trigger correctness or pre-contact completion of configuration in each trial. Future work will add synchronized trigger–contact logging, channel-specific ablations, post-contact verification, and a larger participant sample.

Supplementary Material

Supplementary Tables S1–S3 are supplied as a separate editable supplementary file.

Declarations

Ethics statement

All participants provided written informed consent before participation. The low-risk, non-invasive experimental procedure was conducted in accordance with relevant institutional guidelines; no personally identifiable information was collected, and all data were anonymized.

Image integrity

Figure 1 is a photograph of the actual experimental platform. No generative AI was used to add, remove, replace, or reposition experimental elements. Figures 2–3 were redrawn by the authors using Python/Matplotlib, and Figures 4–9 were generated from experimental data. The authors verified the logic, text, and data mapping in the final figures.

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Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The reproducibility bundle prepared for this study contains de-identified trial-level completion time, master-side trajectory length, outcome data, raw NASA-TLX responses, closed-set vision-validation data, analysis scripts, final figure assets, and available figure-source files. Supplementary Tables S1–S3 are included in the bundle. Before submission, the authors will deposit this bundle in a public repository and replace this statement with the repository name and persistent identifier (DOI or equivalent). Event-level visual/strategy/contact logs, video, and detailed failure-type records were not retained and therefore cannot be retrospectively shared.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI Codex to assist with language editing, consistency checks, and the drafting of Python/Matplotlib code. The authors subsequently reviewed and edited all outputs, verified the technical statements and data mappings, and take full responsibility for the content of the manuscript.

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